

Cod (*Gadus morhua*) in Division 7.a (Irish Sea)

ICES advice on fishing opportunities

ICES advises that when the maximum sustainable yield (MSY) approach and precautionary considerations are applied, there should be zero catch in each of the years 2027 and 2028.

ICES notes the existence of a precautionary management plan, developed and adopted by one of the relevant management authorities for this stock.

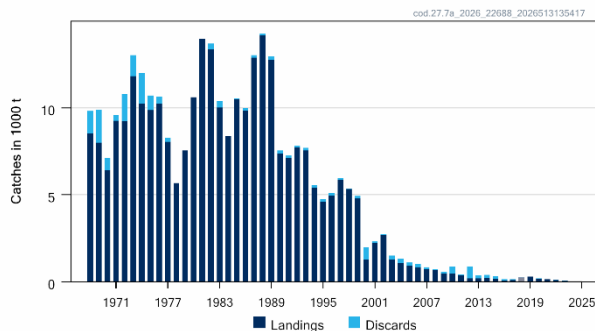
Non-fisheries conservation considerations

Conservation aspects and associated management measures may exist at a national or regional level but were not reviewed by ICES.

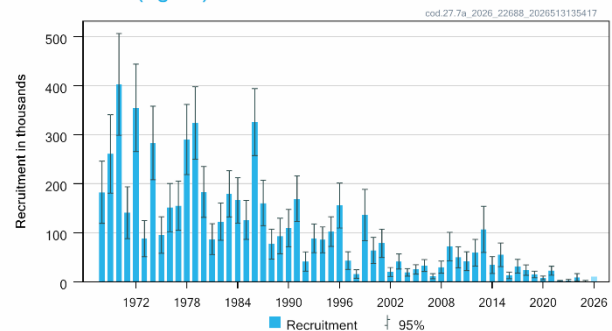
Stock development over time

Fishing pressure on the stock is below F_{MSY} , and spawning-stock size is below MSY $B_{trigger}$, B_{PA} , and B_{lim} .

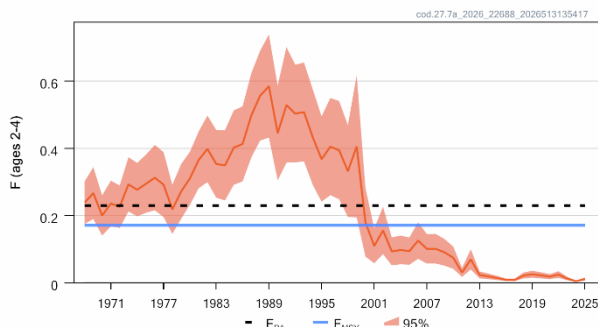
Catches



Recruitment (age 0)



F



SSB

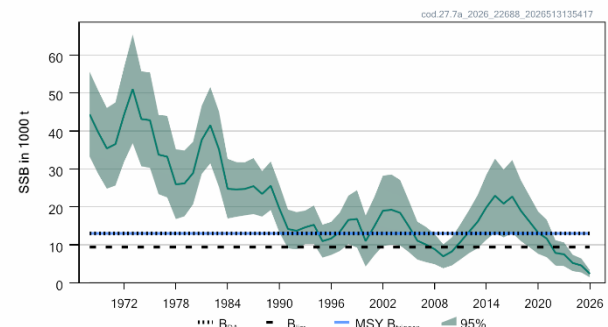


Figure 1 Cod in Division 7.a. Summary of the stock assessment. The assumed recruitment value for 2026 is shaded in a lighter colour.

Catch scenarios

Table 1 Cod in Division 7.a. Values in the forecast and for the interim year.

Variable	Value	Notes
$F_{\text{ages 2-4}} (2026)$	0.0097	Average exploitation pattern (2023–2025), scaled to average $F_{\text{ages 2-4}}$ in 2023–2025
Spawning-stock biomass (SSB; 2027)	3 416	Short-term forecast; in tonnes
$R_{\text{age 0}} (2026, 2027)$	10 077	Geometric mean (2016–2023); in thousands
Total catch (2026)	27	Short-term forecast; in tonnes
Projected landings (2026)	24	Assuming average landing patterns (2023–2025); in tonnes
Projected discards (2026)	3	Assuming average discard patterns (2023–2025); in tonnes

Table 2^{^^} Cod in Division 7.a. Annual catch scenarios. Weights are in tonnes.

Basis	Total catch (2027)	Projected landings (2027)	Projected discards (2027)	$F_{\text{total}} (2027)$	$F_{\text{projected landings}} (2027)$	$F_{\text{projected discards}} (2027)$	Spawning-stock biomass (SSB; 2028)	% SSB change *	% total allowable catch (TAC) change [^]
ICES advice basis									
Maximum sustainable yield (MSY) and precautionary considerations: $F = 0$	0	0	0	0	0	0	3277	-4.1	-100
Other scenarios									
$F_{\text{MSY}} \times \text{SSB}_{2027} / \text{MSY } B_{\text{trigger}}$	112	98	14	0.045	0.039	0.0056	3145	-7.9	-32
EU MAP ^{**} : $F_{\text{MSY lower}} \times \text{SSB}_{2027} / \text{MSY } B_{\text{trigger}}$	93	81	12	0.037	0.032	0.0047	3168	-7.3	-44
EU MAP ^{**} : $F_{\text{MSY lower}}$	338	295	43	0.141	0.123	0.0177	2884	-15.6	105
$F = F_{\text{eco}}$	356	310	45	0.149	0.130	0.0187	2864	-16.5	116
$F_{\text{eco}} \times \text{SSB}_{2027} / \text{MSY } B_{\text{trigger}}$	98	86	12	0.039	0.034	0.0049	3162	-7.4	-41
$F = F_{\text{MSY}}$	404	353	51	0.171	0.150	0.021	2809	-17.8	145
$F = F_{\text{PA}}$	530	462	68	0.23	0.20	0.029	2667	-22	220
$F = F_{2026}$	25	22	3	0.0097	0.0085	0.00122	3248	-4.9	-85
Rollover TAC	165	144	21	0.067	0.058	0.0084	3084	-9.7	0
$\text{SSB}_{2028} = \text{SSB}_{2027}$ ***									
$\text{SSB}_{2028} = B_{\text{lim}}$ ***									
$\text{SSB}_{2028} = B_{\text{PA}}$ ***									
$\text{MSY } B_{\text{trigger}}$ ***									

* SSB 2028 relative to SSB 2027.

** EU multiannual plan (MAP) for the Western Waters (EU, 2019).

*** The SSB_{2027} , B_{lim} , B_{PA} , and $\text{MSY } B_{\text{trigger}}$ options were left blank because none of them can be achieved in 2028, even with zero catch in 2027.

[^] Total catch advice in 2027 relative to the TAC in 2026 (165 tonnes).

^{^^} Calculation of the probability of the stock being below B_{lim} in 2028 is not possible with this deterministic forecast.

The advice for 2027 is the same as for 2026: zero catch. ICES provides zero-catch advice for this stock in 2027 and 2028 because the spawning-stock biomass (SSB) is forecast to remain below B_{lim} under all catch scenarios.

Basis of the advice

Table 3 Cod in Division 7.a. The basis of the advice.

Advice basis	Maximum sustainable yield (MSY) approach
Management plan	ICES is aware of the multiannual management plan (MAP) which has been adopted by the EU for this stock (EU, 2019) and which ICES considers to be precautionary. There is no agreed shared management plan between the EU and the UK for this stock, and ICES provides advice according to ICES MSY approach and precautionary considerations. Catch scenarios consistent with the MAP F_{MSY} ranges are provided.

Quality of the assessment

The survey coverage of the NIGFS-WIBTS-Q4 (2023, 2025), NIGFS-WIBTS-Q1 (2024–2025), and UKFSP (2025) was incomplete because of operational issues, with approximately 25% of the coverage area not sampled annually. Sensitivity analyses indicated that this has minimal impact on the perception of the stock status.

In the 2026 assessment, the value for recruitment in 2024, which was missing because of the absence of the survey, was estimated from the model. This improved model performance and results in a downward revision of the 2024 recruitment estimate in this year's assessment.

Due to the sharp decline in the stock size, fewer fish are available for sampling on both survey and commercial vessels. Furthermore, declining effort within the fleets that mainly target whitefish has led to fewer sampling opportunities. The limited number of individuals available to assess the age structure, maturity-at-age, and weight-at-age just adds uncertainty to the assessment and the retrospective but not to the general perception of the stock.

There is high uncertainty in the stock biomass and forecast SSB as a result of uncertainty in recruitment, stock structure and natural mortality processes.

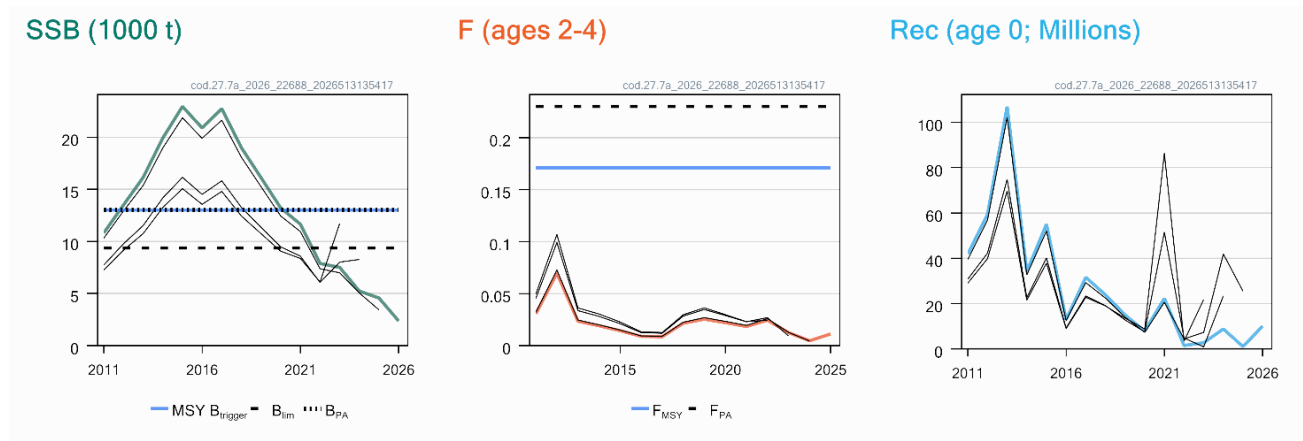


Figure 2 Cod in Division 7.a. Historical assessment results (final-year spawning-stock biomass [SSB] estimate and recruitment assumption included in each line).

Issues relevant for the advice

The assessment and the advice are for Division 7.a, excluding the rectangles 33E2 and 33E3, as cod in these rectangles are considered to be part of the cod.27.7e–k stock (western English Channel and southern Celtic Seas). Landings data used in the cod.27.7a assessment therefore explicitly exclude those reported from rectangles 33E2 and 33E3 (Table 8). Data from these two rectangles are incorporated into the advice for cod.27.7e–k. In 2025, cod landings in 33E2 and 33E3 represented 2.3% of the landings from Division 7.a and constituted 0.7% of the landings from the cod.27.7e–k stock. The spatial limitation of the advice should be considered when setting total allowable catches (TACs) for the two management areas that span the cod.27.7a and 27.7e–k stocks.

Recreational fishing of cod in Division 7.a is known to take place. However, the estimates of recreational catch are uncertain and have not been incorporated into the assessment.

F_{eco} is a dynamic reference point (ICES, 2019, 2023a). F_{eco} is currently set close to F_{MSY} lower (Figure 3).

Mixed-fisheries considerations

Cod in Division 7.a is caught as part of a mixed fishery (ICES, 2026a).

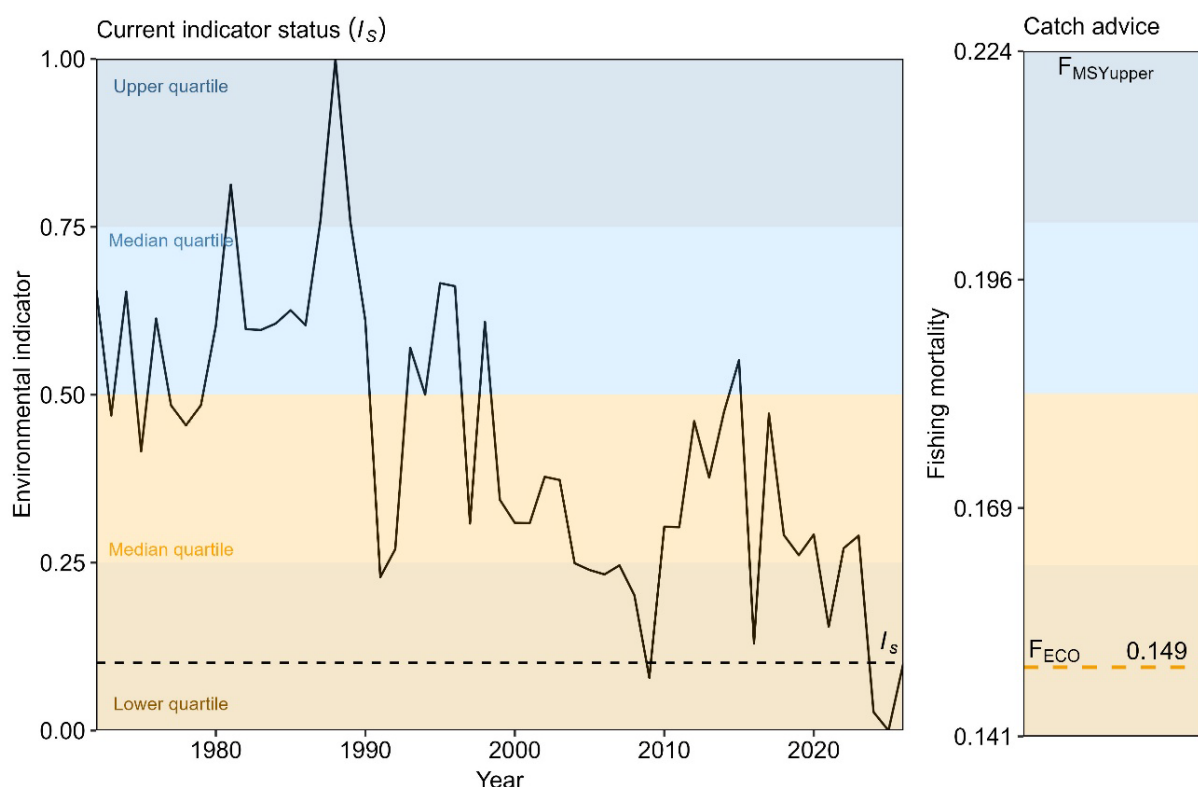


Figure 3 Cod in Division 7.a. Derivation of F_{eco} . Left: time-series of inverted sea-surface temperature (SST; three-year lag) rescaled between zero and one, which informs the status of the indicator (I_s) in 2026 compared with previous years. Right: the status of the indicator determines the placement of the F_{eco} reference point within F_{MSY} ranges (ICES, 2019, 2020, 2023a).

Reference points

Table 4 Cod in Division 7.a. Reference points, values, and their technical basis.

Framework	Reference point	Value	Technical basis	Source
Maximum sustainable yield (MSY) approach	MSY $B_{trigger}$	13012	B_{PA}	ICES (2023a)
	F_{MSY}	0.171	Based on a simulation using model averaged combined stock–recruitment relationship (EqSim)	ICES (2023a)
	F_{eco}	0.149	Ecosystem indicator (I_s); $F_{eco} = F_{MSY\ lower} + ([F_{MSY\ upper} - F_{MSY\ lower}] * I_s)$	ICES (2023a)
Precautionary approach	B_{lim}	9364	Lowest stock spawning biomass (SSB) with above-average recruitment; in tonnes	ICES (2023a)
	B_{PA}	13012	$B_{lim} \times \exp(1.645 \times \sigma)$, $\sigma = 0.20$; in tonnes	ICES (2023a)
	F_{PA}	0.23	F_{POS} ; the F that leads to $SSB > B_{lim}$ with 95% probability	ICES (2023a)
	MAP MSY $B_{trigger}$	13012	MSY $B_{trigger}$; in tonnes	ICES (2023a)
EU Management Plan (MAP)*	MAP B_{lim}	9364	B_{lim} ; in tonnes	ICES (2023a)
	MAP F_{MSY}	0.171	F_{MSY}	ICES (2023a)
	MAP range $F_{MSY\ lower}$	0.141	F at 95% MSY (below F_{MSY}), based on simulation using a model average combined stock–recruitment relationship (EqSim)	ICES (2023a)
	MAP range $F_{MSY\ upper}$	0.22	F at 95 % MSY (above F_{MSY}), based on simulation using model averaged combined stock–recruitment relationship (EqSim)	ICES (2023a)

* EU multiannual plan (MAP) for the Western Waters (EU, 2019).

Basis of the assessment

Table 5 Cod in Division 7.a. Basis of assessment and advice.

ICES stock data category	1 (ICES, 2025)
Assessment type	Age-based analytical model SS3 used in the assessment and FLR in the forecast (ICES, 2026b).
Input data	Northern Ireland Groundfish Survey quarter 1 ([G7144]; 1995–2025) ages 1–4, and quarter 4 ([G7655]; 1995–2025, excluding 2024) age 0 recruitment index, Fishery science partnership survey UKFSP ([B7897]; ages 2–6, 2004–2025, excluding 2014), commercial catches (age composition of landings and discards), annual maturity from the G7144 survey (constant since 2024). Natural mortality estimated from tagging data. Stock weights-at-age were estimated by fitting a von Bertalanffy curve to Q1 survey and catch length-at-age data, accounting for year-class effects; there are not sufficient fish in recent years to estimate length-at-age and weight-at-age.
Discards and bycatch	Discards available since 2007, prior to 2007 are reconstructed (ICES, 2017)
Indicators	Sea surface temperature (Rayner, 2003)
Other information	This stock was benchmarked in 2022 (ICES, 2023b) and reviewed in 2023 (ICES, 2023a)
Working group	Working Group for the Celtic Seas Ecoregion (WGCSE)

History of the advice, catch, and management

Table 6 Cod in Division 7.a. ICES advice, the agreed total allowable catch (TAC), official landings and ICES estimates of landings and discards. Weights are in tonnes.

Year	ICES advice, with single-stock exploitation boundaries since 2004	Catch corresponding to advice	Agreed TAC	Official landings	ICES landings ***	ICES discards
1987	No increase in F; interaction with <i>Nephrops</i>	10300	15000	13200	12894	
1988	No increase in F; interaction with <i>Nephrops</i>	10100	15000	15800	14168	
1989	No increase in F	13400	15000	11300	12751	
1990	F at F_{med} ; TAC	15300	15300	9900	7379	
1991	Stop SSB decline; TAC	6000	10000	7000	7095**	
1992	20% of F(90) ~ 10 000 tonnes	10000	10000	7400	7735**	
1993	F_{med} ~ 10 200 tonnes	10200	11000	5900	7555**	
1994	60% reduction in F	3700	6200	4500	5402**	
1995	50% reduction in F	3900	5800	4500	4587**	
1996	30% reduction in F	5400	6200	5303	4964**	
1997	30% reduction in F	5900	6200	4441	5859**	
1998	No increase in F	6200	7100	4962	5318**	
1999	Reduce F below F_{pa}	4900	5500	2875	4784**	
2000	Lowest possible F	0	2100	1417	1274	
2001	Lowest possible F	0	2100	2026	2252	
2002	Establish recovery plan	-	3200	2715	2695	
2003	Closure of all fisheries for cod	-	1950	1477	1285	
2004	Zero catch	0	2150	1179	1072	
2005	Zero catch	0	2150	967	910	
2006	Zero catch	0	1828	948	840	
2007	Zero catch	0	1462	1117	702	145
2008	Zero catch	0	1199	1224	661	61
2009	Zero catch	0	899	765	468	88
2010	Zero catch	0	674	594	464	386
2011	Zero catch	0	506	485	368	48
2012	Zero catch	0	380	326	198	678
2013	No directed fisheries, minimize bycatch and discards	0	285	281	206	152
2014	No directed fisheries, minimize bycatch and discards	0	182	236	213	184
2015	No directed fisheries, minimize bycatch and discards	0	146	199	161	147
2016	No directed fisheries, minimize bycatch and discards	0	146	122	82	60
2017	Maximum sustainable yield (MSY) approach	0	146	103	84	59
2018	MSY approach	≤ 1073	695	235	215	42
2019	MSY approach	≤ 807	807	205 ^c	295	7
2020	Precautionary approach	≤ 116	257	255	181	25

Year	ICES advice, with single-stock exploitation boundaries since 2004	Catch corresponding to advice	Agreed TAC	Official landings	ICES landings ***	ICES discards
2021	Precautionary approach	≤ 93	206	184	133	4
2022	Precautionary approach	≤ 74	206	128	98	27
2023	MSY approach and precautionary considerations	0	165	76	56	9
2024	MSY approach and precautionary considerations	0	165	16*	11	4
2025	MSY approach and precautionary considerations	0	165	31*	30	2
2026	MSY approach and precautionary considerations	0	165			
2027	MSY approach and precautionary considerations	0				
2028	MSY approach and precautionary considerations	0				

* Preliminary.

** Includes sample-based estimates of landings into three ports.

*** Excludes landings reported from rectangles 33E2 and 33E3 since 2004.

^c Incomplete/missing as a result of part of the data being unavailable under data confidentiality clauses.

History of the catch and landings

Table 7 Cod in Division 7.a. Catch distribution by fleet in 2025 as estimated by ICES, minor differences in total value are due to rounding.

Catch	Landings				Discards			
32 tonnes	<i>Nephrops</i> directed otter trawls 35%	Demersal fish directed otter trawls 55%	Beam trawls 7%	Other gears < 3%	<i>Nephrops</i> directed otter trawls 78%	Demersal fish directed otter trawls 3%	Beam trawls 15%	Other gears 4%
	30 tonnes				2 tonnes			

Table 8 Cod in Division 7.a. History of official landings by country and ICES estimates of landings and discards. Weights are in tonnes; minor differences in total value are due to rounding.

Year	Belgium	France	Ireland	Netherlands	Spain	UK (England Wales, & NI)	UK (Isle of Man)	UK (Scotland)	Total	Landings in rectangles 33E2 & 33E3***	ICES landings	ICES discards
1996	142	148	2476	25	-	2359	27	126	5303		4964**	
1997	183	268	1492	29	-	2370	19	80	4441		5859**	
1998	316	269	1739	20	-	2517	34	67	4962		5318**	
1999	150	n/a	966	5	-	1665	9	80	2875		4784**	
2000	60	53	455	1	-	799	11	38	1417		1274	
2001	283	74	751	-	-	885	1	32	2026		2252	
2002	318	116	1111	-	-	1134	7	29	2715		2695	
2003	183	151	594	-	14	505	7	23	1477		1285	
2004	104	29	380	-	-	646	5	15	1179	108	1072	
2005	115	35	220	-	-	594	n/a	3	967	54	910	
2006	60	18**	275	-	-	589	n/a	6	948	103	840	
2007	67	17**	608	-	-	423	n/a	2	1117	527	702	145
2008	26	3	618**	-	-	543	22	12	1224	558	662	61
2009	19	12	323**	-	-	387	12	12	765	193	466	88
2010	21	1	289	-	-	282	1	-	594	143	464	386
2011	36	3	275	-	-	169	1	-	485	147	365	48
2012	23	1	193	-	-	109	< 1	-	326	85	198	678
2013	13	< 1	160	-	-	107	< 1	-	281	76	206	152
2014	9	< 1	148	-	-	79	< 1	-	236	24	213	184
2015	12	< 1	137	-	-	50	< 1	-	199	39	161	147
2016	3	< 1	84	-	-	35	< 1	-	122	40	82	60
2017	5	< 1	57	-	-	41	< 1	< 1	103	19	84	59
2018	2	< 1	105	-	-	128	< 1	< 1	235	20	215	42
2019	10	< 1	^c	-	-	195	< 1	< 1	205 ^c	37	295	7

Year	Belgium	France	Ireland	Netherlands	Spain	UK (England Wales, & NI)	UK (Isle of Man)	UK (Scotland)	Total	Landings in rectangles 33E2 & 33E3***	ICES landings	ICES discards
2020	10	0	147	-	-	97	< 1	< 1	255	71	181	25
2021	3	0	93	-	-	89	< 1	< 1	184	52	133	4
2022	3	0	55	-	-	70	< 1	< 1	128	30	98	27
2023	1	0	38	-	-	37	< 1	< 1	76	19	56	9
2024*	< 1	0	9	-	-	7	0	<1	16	5	11	4
2025*	1	0	20	-	-	7	< 1	2	31	< 1	30	2

*Preliminary official landings.

** Includes sample-based estimates of landings into ports.

*** Landings in the southern part of Division 7.a (rectangles 33E2 and 33E3) are not included in the assessment and are considered to be part of the cod stock in divisions 7.e–k.

° Incomplete/missing as a result of part of the data being unavailable under data confidentiality clauses.

Summary of the assessment

Table 9 Cod in Division 7.a. Assessment summary. All weights are in tonnes, recruitment in thousands. “Low” and “high” refer to 95% confidence intervals. Landings and discards are ICES estimates, which are used as assessment inputs.

Year	Recruitment age 0			Spawning stock biomass (SSB)			Landings	Discards **	F		
	Low	Value	High	Low	Value	High			Low	Value	High
1968	118632	182030	245751	33155	44306	55514	8541	1285	0.21	0.28	0.42
1969	180475	260144	340219	28572	39543	50569	7991	1898	0.23	0.31	0.46
1970	298529	402050	506099	24669	35283	45951	6426	708	0.173	0.23	0.34
1971	87805	140587	193638	25520	36445	47426	9246	363	0.21	0.27	0.41
1972	265458	354282	443560	31779	44206	56697	9234	1546	0.200	0.26	0.39
1973	51088	87839	124776	36682	50975	65341	11819	1222	0.26	0.34	0.51
1974	207912	282671	357812	30625	43097	55633	10251	1749	0.24	0.32	0.48
1975	58092	95207	132511	30216	42773	55394	9863	857	0.26	0.34	0.50
1976	101707	151016	200576	23299	33713	44180	10247	381	0.27	0.36	0.52
1977	105011	154828	204899	22446	33179	43966	8054	201	0.24	0.34	0.48
1978	218382	289877	361737	16747	25926	35152	5662	0	0.180	0.25	0.36
1979	249527	323439	397728	17486	26140	34838	7548	0	0.23	0.31	0.46
1980	131104	183024	235209	20668	28897	37169	10599	0	0.28	0.36	0.55
1981	55340	86896	118613	28652	37617	46628	13958	0	0.34	0.42	0.67
1982	84551	122322	160286	31451	41490	51581	13381	313	0.36	0.46	0.71
1983	132207	179279	226591	25188	35121	45104	10015	372	0.31	0.41	0.61
1984	119750	165898	212282	16907	24761	32656	8383	2	0.30	0.41	0.59
1985	86270	126208	166349	17376	24569	31798	10483	61	0.36	0.47	0.70
1986	257358	325362	393713	17743	24721	31735	9852	154	0.37	0.48	0.72
1987	114293	160380	206702	18082	25444	32845	12894	128	0.45	0.58	0.89
1988	46328	76642	107111	17422	23400	29407	14168	109	0.51	0.65	1.01
1989	56786	92934	129267	19157	25534	31944	12751	202	0.53	0.68	1.04
1990	71244	109247	147444	13251	19541	25863	7379	159	0.38	0.52	0.74
1991	122100	168766	215670	9041	14141	19266	7095	163	0.44	0.61	0.87
1992	21361	40943	60624	8762	13657	18578	7735	98	0.44	0.59	0.87
1993	58592	87863	117283	10170	14557	18965	7555	155	0.44	0.59	0.88
1994	58236	84657	111212	10119	15174	20254	5402	142	0.36	0.51	0.72
1995	71531	101698	132019	6523	10854	15207	4587	166	0.30	0.43	0.60
1996	108248	153906	199797	7181	11522	15886	4964	140	0.33	0.48	0.65
1997	24771	42588	60496	8184	13225	18292	5859	120	0.32	0.46	0.62
1998	6807	15438	24113	9955	16322	22720	5318	29	0.25	0.39	0.50
1999	81681	133914	186413	8988	16535	24120	4784	159	0.27	0.48	0.53
2000	36586	63084	89717	4022	10722	17456	1274	699	0.111	0.21	0.22
2001	48364	77015	105813	6864	14526	22226	2252	64	0.126	0.178	0.25
2002	10431	19614	28844	9352	18560	27816	2695	46	0.182	0.25	0.36

Year	Recruitment age 0			Spawning stock biomass (SSB)			Landings	Discards **	F		
	Low	Value	High	Low	Value	High			Low	Value	High
2003	25425	40811	56276	9575	18816	28105	1285	215	0.110	0.151	0.22
2004	11120	18835	26590	9399	18022	26689	1072	254	0.115	0.158	0.23
2005	15210	24724	34286	7784	14565	21381	910	204	0.112	0.153	0.22
2006	20429	32528	44689	5874	10854	15859	840	185	0.149	0.20	0.29
2007	5860	11046	16258	5214	9845	14499	702	145	0.120	0.163	0.24
2008	16848	29239	41693	4709	8678	12667	662	61	0.120	0.164	0.24
2009	41462	70512	99711	3692	6831	9987	466	88	0.107	0.147	0.21
2010	27403	48860	70425	4394	8080	11784	464	386	0.088	0.121	0.174
2011	21826	41064	60401	5954	10593	15255	365	48	0.036	0.050	0.071
2012	31167	58147	85264	7566	13227	18916	198	678	0.083	0.113	0.163
2013	57721	104700	151918	8789	15797	22841	206	152	0.028	0.038	0.055
2014	16122	33219	50404	10944	19500	28100	213	184	0.023	0.031	0.045
2015	29855	53878	78023	12723	22458	32242	161	147	0.0170	0.023	0.034
2016	6002	12662	19357	11505	20422	29385	82	60	0.0107	0.0145	0.021
2017	16870	30803	44807	12620	22218	31864	84	59	0.0103	0.0139	0.020
2018	12781	23320	33913	10564	18561	26598	215	42	0.026	0.035	0.051
2019	7899	14579	21293	8975	15731	22521	295	7	0.031	0.041	0.061
2020	3911	7785	11679	7320	12884	18476	181	25	0.027	0.036	0.053
2021	13890	22875	31906	6439	11360	16305	133	4	0.022	0.030	0.044
2022	364	1491	2623	4371	7689	11023	98	27	0.029	0.039	0.058
2023	720	3021	5335	4269	7387	10521	56	9	0.0149	0.0199	0.029
2024	2200	10068	17976	3026	5180	7346	11	4	0.0055	0.0074	0.0109
2025	0	10036	27314	2437	4068	5708	30	2	0.0144	0.0190	0.028
2026		10077*		1467	2424	3386					

*Geometric mean 2016–2023.

** Discards prior to 2007 are reconstructed.

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Recommended citation: ICES. 2026. Cod (*Gadus morhua*) in Division 7.a (Irish Sea). *In* Report of the ICES Advisory Committee, 2026. ICES Advice 2026, cod.27.7a. <https://doi.org/10.17895/ices.advice.30932033>